Grade 8 Name Answer Key
Unit 4 Date____________________
Lesson 5 Practice Period____________________

Hadley is evaluating the expression $\frac{(4.5)^{-5}\cdot\ (4.5)^{0}}{(4.5^{3})^{-2}}$.

1. Which of the expression(s) show the correct first step in evaluating the expression above?

A. $\frac{(4.5)^{-5}}{(4.5^{3})^{-2}}$ B. $\frac{(4.5)^{0}}{(4.5)^{-6}}$ C. $\frac{(4.5)^{-5}\cdot\ (4.5)^{0}}{(4.5)^{-6}}$ D. $\frac{(4.5)^{-5}\cdot\ (4.5)^{0}}{(4.5)^{6}}$

2. Which exponent law(s) was applied in the first step?

- ☐ Zero Exponent
- ☐ Identify Exponent
- ☐ Negative Exponent
- ☒ Power of a Power
- ☒ Product of Powers
- ☐ Power of a Product
- ☐ Quotient of Powers
- ☐ Power of a Quotient

3. Evaluate the expression $\frac{(4.5)^{-5}\cdot\ (4.5)^{0}}{(4.5^{3})^{-2}}$.

$\frac{(4.5)^{-5}}{(4.5)^{-6}} = (4.5)^{-5-(-6)} = (4.5)^{1} = 4.5$

Evaluate the numerical expressions. Show all the steps using the laws of exponents.

4. $\left(\frac{3^{4}}{3^{-2}}\right)^{-3}$

$\frac{(3)^{-12}}{3^{6}} = 3^{-12-6} = \frac{1}{3^{18}} = \frac{1}{387{,}420{,}489}$

5. Which exponent law was applied first when evaluating question 4?

Answers will vary. Power of a Power, Quotient of Powers.

6. $\frac{(3 \cdot 8)^3}{3^2 \cdot 3^2} = \frac{3^3 \cdot 8^3}{3^2 \cdot 3^2} = \frac{3^3 \cdot 8^3}{3^4} = 3^{3-4}\ 8^3 = 3^{-1} \cdot 8^3 = \frac{1}{3} \cdot 8^3 = \frac{512}{3}$

7. Which exponent law was applied first when evaluating question 6?

Answers will vary. Power of a Power, Product of Powers, Power of a Product.

8. $\frac{\left(\frac{2}{3}\right)^{-5} \cdot \left(\frac{2}{3}\right)^0}{\left(\frac{2}{3}\right)^{-2}} = \frac{\left(\frac{2}{3}\right)^{-5}}{\left(\frac{2}{3}\right)^{-2}} = \left(\frac{2}{3}\right)^{-5-(-2)} = \left(\frac{2}{3}\right)^{-3} = \left(\frac{3}{2}\right)^3 = \frac{27}{8}$

9. Which exponent law was applied first when evaluating question 8?

Answers will vary. Zero Exponent, Product of Powers.

10. Lauren evaluated the expression $\frac{(6^{-1}) \cdot (6^5)^0}{6^2}$. Her work is shown below. Circle the step where she made an error and describe the mistake she made.

Problem	Describe the Mistake
$\frac{(6^{-1}) \cdot (6^5)^0}{6^2}$ $\frac{(6^{-1}) \cdot 6^5}{6^2}$ $\frac{6^4}{6^2}$ 6^3 216	Lauren added the exponents of 5 and 0. She should have multiplied them to have a power of zero and not five.